Software Requirements Specification (SRS)

Otithi

Authentic Bangladeshi Local Hospitality Platform

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1 Introduction

1.1 Background

Otithi represents a significant advancement in Bangladesh's digital hospitality landscape. The name "Otithi," which means "guest" in Bengali, reflects the traditional Bangladeshi value of *atithi devo bhava*, emphasizing the importance of treating guests with the utmost respect and honor. This platform is designed to bridge the technological gaps in Bangladesh's accommodation market while acknowledging the cultural significance of hospitality in Bengali life.

This initiative is timely, addressing a pressing market demand. The tourism industry in Bangladesh has seen impressive growth in recent years, with domestic and international travel increasing by 15–20% annually. However, the accommodation infrastructure has primarily remained conventional and inadequate. International platforms like Airbnb have a minimal presence in Bangladesh, and local options often lack the comprehensive features that modern travelers expect.

Otithi is built on a robust technological framework incorporating advanced web development techniques. It utilizes the Python Flask framework with a MySQL database, alongside a frontend developed using HTML5, CSS3, and JavaScript with the Bootstrap framework. Security features include Flask-Login for authentication, CSRF protection, and password hashing, ensuring a secure user experience. The development stack is based on XAMPP, which allows for strong performance and scalability while effectively aligning with Bangladesh's digital environment.

By addressing the critical market need for improved accommodation options, Otithi aims to enhance the experience of travelers in Bangladesh. With the rapid growth of the tourism industry, this platform represents an essential step forward in modernizing the country's hospitality sector.

1.2 Motivation

The motivation for developing the Otithi platform is driven by several interconnected factors that create a strong business and social case:

Economic Impact: The platform allows property owners to monetize previously unused spaces, which could potentially increase accommodation capacity in Bangladesh by 30–40%. This not only generates new revenue streams for homeowners but also enhances the country's appeal as a tourist destination. The resulting economic activity supports local businesses through increased visitor spending, creates employment opportunities in hospitality and property management, and contributes to the growth of the digital economy.

Cultural Exchange: Otithi promotes meaningful connections between travelers and local hosts, enabling cultural immersion that transcends traditional tourism. The platform encourages Bengali-English language exchange, showcases the renowned warmth and hospitality of the Bangladeshi people to international visitors, and bridges cultural gaps through shared accommodations. This cultural exchange fosters mutual understanding and highlights Bangladesh's rich heritage.

Technological Advancement: The platform modernizes Bangladesh's accommodation booking system, aligning it with international standards. It offers a mobile-first booking experience that caters to the country's high smartphone penetration rate—over

80% of internet users access the web via mobile devices. Additionally, the platform facilitates secure online transactions in Bangladeshi Taka (BDT), addressing a significant gap in the current market, and generates valuable insights for tourism planning and development.

1.3 Problem Statement

The Otithi platform addresses several significant market gaps and user pain points:

1. **Limited Accommodation Options:** Traditional hotels and guesthouses often fall short of providing the authentic local experiences that modern travelers increasingly seek. The market currently lacks a diverse range of accommodation types and unique experiences.

2. **Trust and Safety Concerns:** There is no standardized platform for verifying hosts and properties, which creates uncertainty for both hosts and guests. This absence of trust mechanisms significantly hinders the growth of peer-to-peer accommodation sharing.

3. **Booking Complexity:** Current booking processes typically involve manual coordination, phone calls, and cash transactions, which lack the convenience and transparency offered by online platforms. This creates friction in the user experience.

4. **Payment Challenges:** The lack of integrated payment systems supporting the Bangladeshi Taka (BDT) presents significant barriers for both domestic and international users. Many international platforms do not support local currency, while local solutions often do not provide the security and convenience expected by modern users.

5. **Discovery Difficulty:** There is no centralized platform for discovering unique accommodations across Bangladesh. Travelers must rely on scattered information sources, which makes trip planning inefficient and can lead to missing out on exceptional experiences.

6. **Communication Barriers:** There are limited tools for host-guest interaction in the local context, particularly with respect to language preferences and cultural nuances specific to Bangladesh.

The current market landscape reveals several critical gaps: no dedicated accommodation sharing platform specifically designed for Bangladesh, absence of local currency (BDT) integration in international platforms, limited support for Bengali language preferences, lack of Bangladesh-specific features like local payment methods, and absence of cultural context in accommodation experiences.

1.4 Objectives

1.4.1 Primary Objectives

1. **Platform Development:** Create a comprehensive, user-friendly accommodation rental platform that serves as the definitive solution for accommodation sharing in Bangladesh, incorporating modern web technologies and user experience principles.

2. **User Acquisition:** Attract 1,000+ verified hosts and 5,000+ registered guests within the first year of operation, establishing a critical mass necessary for platform sustainability and growth.

3. **Booking Facilitation:** Process 10,000+ successful bookings annually, creating a robust transaction volume that demonstrates platform viability and generates sustainable revenue through service fees.

4. **Revenue Generation:** Achieve a sustainable business model through service fees, premium features, and strategic partnerships, ensuring long-term platform viability and continued development.

5. **Market Expansion:** Cover major cities and tourist destinations across Bangladesh, including Dhaka, Chittagong, Sylhet, Cox's Bazar, and emerging tourist areas, ensuring comprehensive market coverage.

1.4.2 Secondary Objectives

1. Foster a trusted, engaged community of hosts and guests through robust verification systems, transparent review mechanisms, and community guidelines that promote positive interactions.

2. Maintain high quality standards through comprehensive review systems, host verification processes, and continuous quality monitoring to ensure user satisfaction and platform reputation.

3. Implement cutting-edge features for enhanced user experience, including AI-powered recommendations, advanced search capabilities, and seamless mobile experiences that exceed user expectations.

4. Collaborate with tourism boards, local businesses, and government entities to promote responsible tourism and contribute to Bangladesh's tourism development goals.

5. Promote responsible tourism practices and local economic development by encouraging sustainable accommodation practices and supporting local businesses through increased visitor spending.

2 System Overview

The Otithi platform represents a sophisticated, multi-layered system designed to handle the complexities of accommodation sharing while maintaining simplicity for end users. The architecture follows industry best practices for scalability, security, and user experience.

2.1 System Architecture

The platform implements a robust three-tier architecture that separates concerns and ensures maintainability:

Presentation Layer: This layer provides responsive web interfaces that adapt seamlessly across devices, from desktop computers to mobile phones. It includes role-based user interfaces that dynamically adjust based on user permissions (Guest, Host, Admin), and RESTful API endpoints that enable future mobile application development and third-party integrations.

Business Logic Layer: The core application server built on Flask handles all business operations including user authentication and authorization, comprehensive booking workflow management, secure payment processing with multiple gateway support, and sophisticated communication systems for host-guest interactions.

Data Layer: The MySQL database provides reliable data storage with optimized query performance, comprehensive file storage for images and documents, automated backup mechanisms with disaster recovery capabilities, and analytics capabilities for business intelligence and user behavior analysis.

2.2 Core System Components

The platform consists of several interconnected components that work together to provide a seamless user experience:

User Management System: Handles user registration with email verification, secure authentication using industry-standard protocols, comprehensive profile management with photo uploads and verification, and role-based access control ensuring appropriate permissions for different user types.

Property Management System: Enables hosts to create detailed property listings with rich media content, manage property details and availability calendars, and participate in a verification workflow that ensures quality and safety standards.

Booking Engine: Provides advanced search functionality with multiple filter options, real-time availability checking, secure booking process with confirmation workflows, and comprehensive booking management for both hosts and guests.

Communication System: Facilitates in-platform messaging between hosts and guests, automated notification systems for booking updates and important events, and a comprehensive review and rating system that builds trust and quality standards.

Administrative Dashboard: Provides platform administrators with comprehensive tools for user management, content moderation, analytics and reporting, and system configuration management.

2.3 Technology Stack

The platform leverages a carefully selected technology stack that balances performance, security, and maintainability:

Backend: Python 3.13 with Flask framework provides robust server-side functionality, MySQL database ensures reliable data storage and retrieval, and comprehensive security measures including HTTPS/TLS encryption and CSRF protection.

Frontend: HTML5, CSS3, and JavaScript with Bootstrap 5.1.3 framework create responsive, modern user interfaces, while Google Maps API integration provides location-based services and enhanced user experience.

Development Environment: XAMPP development environment ensures consistent development and testing across different systems, while comprehensive version control and deployment pipelines ensure reliable software delivery.

3 System Life Cycle

The development of the Otithi platform follows a structured, five-phase approach designed to ensure quality, manage risks, and deliver value incrementally. This methodology balances rapid development with thorough testing and validation.

3.1 Development Phases

Phase 1: Foundation (Months 1-2) - This critical foundation phase establishes the core architecture, database design with optimized schemas, user authentication systems with security best practices, and basic user interface frameworks. This phase creates the solid foundation upon which all subsequent features will be built.

Phase 2: Core Features (Months 3-4) - Building upon the foundation, this phase implements property management capabilities, comprehensive booking systems, secure file upload mechanisms, and administrative panel functionality. These features represent the essential functionality required for basic platform operation.

Phase 3: Enhanced Features (Months 5-6) - This phase introduces advanced capabilities including payment integration with multiple gateway support, sophisticated messaging systems, comprehensive review and rating mechanisms, and mobile optimization for enhanced user experience across devices.

Phase 4: Advanced Features (Months 7-8) - The platform reaches maturity with analytics and reporting capabilities, comprehensive API development for future integrations, security hardening and penetration testing, and performance optimization for scalability and user experience.

Phase 5: Launch Preparation (Months 9-10) - The final phase focuses on comprehensive testing across multiple environments, complete documentation for users and administrators, production deployment with monitoring systems, and strategic marketing initiatives to drive user adoption.

3.2 Maintenance and Evolution

Post-launch maintenance follows a structured approach ensuring continuous improvement and system reliability:

Regular Updates: We provide monthly feature updates and bug fixes based on user feedback and system monitoring, ensuring the platform stays aligned with user needs and technological advancements.

Security Maintenance: Immediate addressing of security vulnerabilities through continuous monitoring, regular security audits, and rapid response to emerging threats.

Infrastructure Scaling: Proactive infrastructure scaling based on user growth patterns, ensuring consistent performance and user experience as the platform expands.

Continuous Improvement: Ongoing enhancement based on user feedback, usage analytics, and market trends, ensuring the platform remains competitive and valuable to users.

4 Functional Requirements

4.1 List of Features with Brief Descriptions

The Otithi platform encompasses a comprehensive set of features designed to address all aspects of the accommodation sharing experience:

1. **User Registration & Authentication:** Secure user account creation with email verification, multi-factor authentication options, and robust password policies ensuring account security and user trust.

2. **Profile Management:** Complete user profiles with photo uploads, verification badges, and comprehensive information sharing that builds trust between hosts and guests.

3. **Property Listing Creation:** Intuitive tools for hosts to create detailed property listings with high-quality photos, comprehensive descriptions, and accurate pricing information.

4. **Property Management:** Comprehensive tools for editing listing details, managing availability calendars, and controlling listing visibility to optimize booking opportunities.

5. **Search & Discovery:** Advanced search functionality with multiple filters including location, dates, price ranges, amenities, and user preferences, enabling efficient property discovery.

6. **Booking System:** Complete reservation process with real-time availability checking, instant confirmation, and comprehensive booking management for both hosts and guests.

7. **Payment Processing:** Secure transaction processing in Bangladeshi **Taka** (BDT) with multiple payment gateway support, ensuring convenient and secure financial transactions.

8. **Messaging System:** In-platform communication system enabling direct host-guest interaction, automated notifications, and message history for comprehensive communication tracking.

9. **Review & Rating:** Post-stay review system with 5-star ratings, detailed feedback mechanisms, and moderation systems ensuring quality and authenticity of reviews.

10. **Administrative Dashboard:** Comprehensive platform management tools including user management, content moderation, analytics reporting, and system configuration capabilities.

11. **Notification System:** Multi-channel notification system including email and in-app notifications for booking updates, messages, and important platform announcements.

12. **Mobile Optimization:** Responsive design ensuring optimal user experience across all devices, from desktop computers to mobile phones and tablets.

13. **AI Tour Guide Chat:** Intelligent virtual assistant providing real-time support, property information, and booking assistance through natural language conversation interface.

14. **Favorites & Wishlist:** User-curated collections of preferred properties with save functionality for future reference and easy access to desired accommodations.

15. **Map Integration:** Google Maps API integration for location-based property discovery, route planning, and geographical context for accommodation choices.

16. **Photo Gallery Management:** Advanced image handling with multiple photo uploads, gallery organization, and image optimization for fast loading and better user experience.

17. **Availability Calendar:** Interactive calendar interface for hosts to manage booking availability and for guests to view property availability in real-time.

18. **Multi-language Support:** Bengali and English language interface with cultural context considerations for local and international users.

19. **File Upload System:** Secure document and image upload capabilities with validation, compression, and storage optimization for property listings and user verification.

20. **Email Verification System:** Automated email confirmation process for user registration and important account activities ensuring platform security and user authenticity.

21. **Password Security:** Advanced password hashing with bcrypt, strong password policies, and secure password reset functionality for enhanced account protection.

22. **CSRF Protection:** Cross-site request forgery protection on all state-changing operations ensuring data integrity and security across the platform.

23. **Session Management:** Secure user session handling with automatic timeout, secure cookie management, and protection against session hijacking attacks.

24. **Error Handling:** Comprehensive error management system with user-friendly error messages, logging, and graceful degradation for improved user experience.

25. **Data Backup & Recovery:** Automated database backup systems with disaster recovery capabilities ensuring data protection and business continuity.

26. **Performance Optimization:** Database query optimization, caching mechanisms, and image compression for fast loading times and optimal platform performance.

4.2 Database Schema Overview

The platform utilizes a comprehensive database schema with the following key tables:

Users Table Stores user account information, authentication details, and profile data.

Column	Data Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique user identifier
username	VARCHAR(50)	UNIQUE, NOT NULL	User's login username
email	VARCHAR(100)	UNIQUE, NOT NULL	User's email address
password_hash	VARCHAR(255)	NOT NULL	Encrypted password
first_name	VARCHAR(50)	NOT NULL	User's first name
last_name	VARCHAR(50)	NOT NULL	User's last name
phone	VARCHAR(20)	NULL	Contact phone number
profile_picture	VARCHAR(255)	NULL	Profile image path
user_type	ENUM('guest','host','admin')	NOT NULL	User role in system
is_verified	BOOLEAN	DEFAULT FALSE	Email verification status
is_active	BOOLEAN	DEFAULT TRUE	Account active status
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Account creation date
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP ON UPDATE	Last update timestamp

Table 1: Users Table Schema

Properties Table Contains property listings, descriptions, pricing, and availability information.

Column	Data Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique property identifier
host_id	INT	FOREIGN KEY (Users.id), NOT NULL	Property owner reference
title	VARCHAR(200)	NOT NULL	Property listing title
description	TEXT	NOT NULL	Detailed property description
property_type	ENUM('apartment','house','villa','room')	NOT NULL	Type of accommodation
address	VARCHAR(500)	NOT NULL	Property location address
city	VARCHAR(100)	NOT NULL	City name
state	VARCHAR(100)	NOT NULL	State/Division name
zip_code	VARCHAR(20)	NULL	Postal/ZIP code
latitude	DECIMAL(10,8)	NULL	GPS latitude coordinate
longitude	DECIMAL(11,8)	NULL	GPS longitude coordinate
price_per_night	DECIMAL(10,2)	NOT NULL	Nightly rate in BDT
max_guests	INT	NOT NULL	Maximum guest capacity
bedrooms	INT	NOT NULL	Number of bedrooms
bathrooms	INT	NOT NULL	Number of bathrooms
amenities	JSON	NULL	Available amenities list
is_available	BOOLEAN	DEFAULT TRUE	Property availability status
is_verified	BOOLEAN	DEFAULT FALSE	Admin verification status
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Listing creation date
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP ON UPDATE	Last update timestamp

Table 2: Properties Table Schema

Bookings Table Manages reservation data, dates, payment status, and guest information.

Column	Data Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique booking identifier
property_id	INT	FOREIGN KEY (Properties.id), NOT NULL	Property reference
guest_id	INT	FOREIGN KEY (Users.id), NOT NULL	Guest user reference
host_id	INT	FOREIGN KEY (Users.id), NOT NULL	Host user reference
check_in_date	DATE	NOT NULL	Arrival date
check_out_date	DATE	NOT NULL	Departure date
total_nights	INT	NOT NULL	Duration of stay
total_amount	DECIMAL(10,2)	NOT NULL	Total booking cost
status	ENUM('pending','confirmed','cancelled','completed')	DEFAULT 'pending'	Booking status
payment_status	ENUM('pending','paid','refunded')	DEFAULT 'pending'	Payment status
special_requests	TEXT	NULL	Guest special requirements
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Booking creation date
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP ON UPDATE	Last update timestamp

Table 3: Bookings Table Schema

Messages Table Stores communication between hosts and guests.

Column	Data Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique message identifier
sender_id	INT	FOREIGN KEY (Users.id), NOT NULL	Message sender reference
receiver_id	INT	FOREIGN KEY (Users.id), NOT NULL	Message recipient reference
booking_id	INT	FOREIGN KEY (Bookings.id), NULL	Related booking reference
message_text	TEXT	NOT NULL	Message content
is_read	BOOLEAN	DEFAULT FALSE	Message read status
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Message timestamp

Table 4: Messages Table Schema

Reviews Table Contains user ratings, feedback, and review moderation data.

Column	Data Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique review identifier
booking_id	INT	FOREIGN KEY (Bookings.id), NOT NULL	Booking reference
reviewer_id	INT	FOREIGN KEY (Users.id), NOT NULL	Review author reference
property_id	INT	FOREIGN KEY (Properties.id), NOT NULL	Property reference
rating	INT	CHECK (1-5), NOT NULL	1-5 star rating
review_text	TEXT	NOT NULL	Detailed review content
is_approved	BOOLEAN	DEFAULT FALSE	Admin approval status
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Review creation date
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP ON UPDATE	Last update timestamp

Table 5: Reviews Table Schema

Payments Table Tracks financial transactions and payment processing details.

Column	Data Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique payment identifier
booking_id	INT	FOREIGN KEY (Bookings.id), NOT NULL	Booking reference
payer_id	INT	FOREIGN KEY (Users.id), NOT NULL	Payment sender reference
payee_id	INT	FOREIGN KEY (Users.id), NOT NULL	Payment recipient reference
amount	DECIMAL(10,2)	NOT NULL	Payment amount in BDT
payment_method	ENUM('bkash','nagad','rocket','card','bank')	NOT NULL	Payment method used
transaction_id	VARCHAR(100)	UNIQUE, NULL	External transaction reference
status	ENUM('pending','completed','failed','refunded')	DEFAULT 'pending'	Payment status
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Payment timestamp

Table 6: Payments Table Schema

Images Table Manages property photos and user profile pictures.

Column	Data Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique image identifier
property_id	INT	FOREIGN KEY (Properties.id), NULL	Property reference
user_id	INT	FOREIGN KEY (Users.id), NULL	User reference
image_path	VARCHAR(500)	NOT NULL	File storage path
image_type	ENUM('property','profile','document')	NOT NULL	Image category
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Upload timestamp

Table 7: Images Table Schema

Notifications Table Stores system notifications and user alerts.

Column	Data Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique notification identifier
user_id	INT	FOREIGN KEY (Users.id), NOT NULL	Recipient user reference
title	VARCHAR(200)	NOT NULL	Notification title
message	TEXT	NOT NULL	Notification content
type	ENUM('booking','message','payment','system')	NOT NULL	Notification category
is_read	BOOLEAN	DEFAULT FALSE	Read status
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Notification timestamp

Table 8: Notifications Table Schema

4.3 Feature Table

Feature ID	Feature Name	User Type	Priority	Description
F001	User Registration	All	High	Account creation with verification
F002	User Authentication	All	High	Secure login and session management
F003	Profile Management	All	High	Complete user profile customization
F004	Property Listing	Host	High	Create and manage property listings
F005	Property Search	Guest	High	Advanced search with filters
F006	Booking System	Guest	High	Complete reservation process
F007	Payment Processing	Guest	High	Secure BDT transactions
F008	Messaging	Host, Guest	Medium	In-platform communication
F009	Review System	Guest	Medium	Post-stay feedback and ratings
F010	Admin Dashboard	Admin	High	Platform management tools
F011	Notifications	All	Medium	Multi-channel alerts
F012	Mobile Interface	All	High	Responsive design
F013	Availability Calendar	Host	Medium	Manage booking availability
F014	Photo Upload	Host	Medium	Property image management
F015	Map Integration	All	Low	Location-based services
F016	Analytics	Admin	Medium	Platform performance metrics
F017	Security Features	All	High	Multi-layer security
F018	Backup System	Admin	High	Data protection

Table 9: Comprehensive Feature Table

4.4 AI Chat Interface Integration

The Otithi platform features an integrated AI-powered virtual tour guide chat interface that provides real-time assistance to users. This interactive chat system serves as a comprehensive support tool, offering personalized guidance and information about properties, locations, and platform features.

4.4.1 Otithi Tour Guide Chat Interface

1. **Virtual Assistant Integration:** The platform includes an intelligent chat interface titled "Otithi Tour guide" that serves as a virtual concierge for users, providing instant support and information.
2. **AI-Powered Responses:** The chat system utilizes advanced AI technology to understand user queries and provide contextual, helpful responses about properties, booking processes, and platform features.
3. **Real-Time Interaction:** Users can engage in natural conversations with the AI assistant, receiving immediate assistance without waiting for human support agents.
4. **Website Knowledge Integration:** The AI assistant can access and reference website content to provide accurate, up-to-date information about properties, amenities, and platform policies.

4.4.2 Chat Interface Features

1. **User-Friendly Design:** Clean, intuitive chat interface with a dark green header and professional styling that matches the platform's visual identity.
2. **Message History:** Persistent chat history that maintains conversation context and allows users to reference previous interactions.
3. **Character Limit Management:** 2000-character input limit with real-time character counting to ensure messages remain concise and manageable.
4. **Feedback System:** Built-in thumbs-up and thumbs-down rating system allowing users to provide feedback on AI responses for continuous improvement.
5. **Timestamp Display:** Real-time timestamp indicators showing message timing for better conversation tracking.

4.4.3 AI Assistant Capabilities

- **Property Information:** Provides detailed information about available properties, amenities, and booking availability
- **Booking Assistance:** Guides users through the booking process, explaining steps and requirements
- **Location Guidance:** Offers information about tourist attractions, local amenities, and area highlights

- **Platform Support:** Answers questions about account management, payment processes, and platform policies
- **Multilingual Support:** Capable of handling queries in multiple languages including Bengali and English

4.4.4 Technical Implementation

The chat interface is implemented using:

- **Frontend Technologies:** HTML5, CSS3, and JavaScript for responsive chat interface
- **AI Integration:** Advanced language processing models for natural conversation understanding
- **Real-Time Communication:** WebSocket technology for instant message delivery and response
- **Content Management:** Integration with platform database for accessing property and user information
- **Responsive Design:** Mobile-optimized interface ensuring consistent experience across all devices

4.4.5 User Experience Benefits

- **24/7 Availability:** Round-the-clock assistance without human agent limitations
- **Instant Response:** Immediate answers to common questions and concerns
- **Personalized Support:** Context-aware responses based on user behavior and preferences
- **Reduced Support Load:** Decreases the need for human customer service intervention
- **Enhanced User Engagement:** Interactive support system that improves overall platform satisfaction

4.4.6 Continuous Learning and Improvement

1. **Feedback Integration:** User ratings and feedback are analyzed to improve AI response quality and accuracy
2. **Conversation Analytics:** Chat interactions are analyzed to identify common user needs and optimize responses
3. **Content Updates:** AI knowledge base is continuously updated with new property information and platform features
4. **Performance Monitoring:** Response accuracy and user satisfaction metrics are tracked for ongoing optimization

5 System Specification Diagrams

5.1 Use Case Diagram

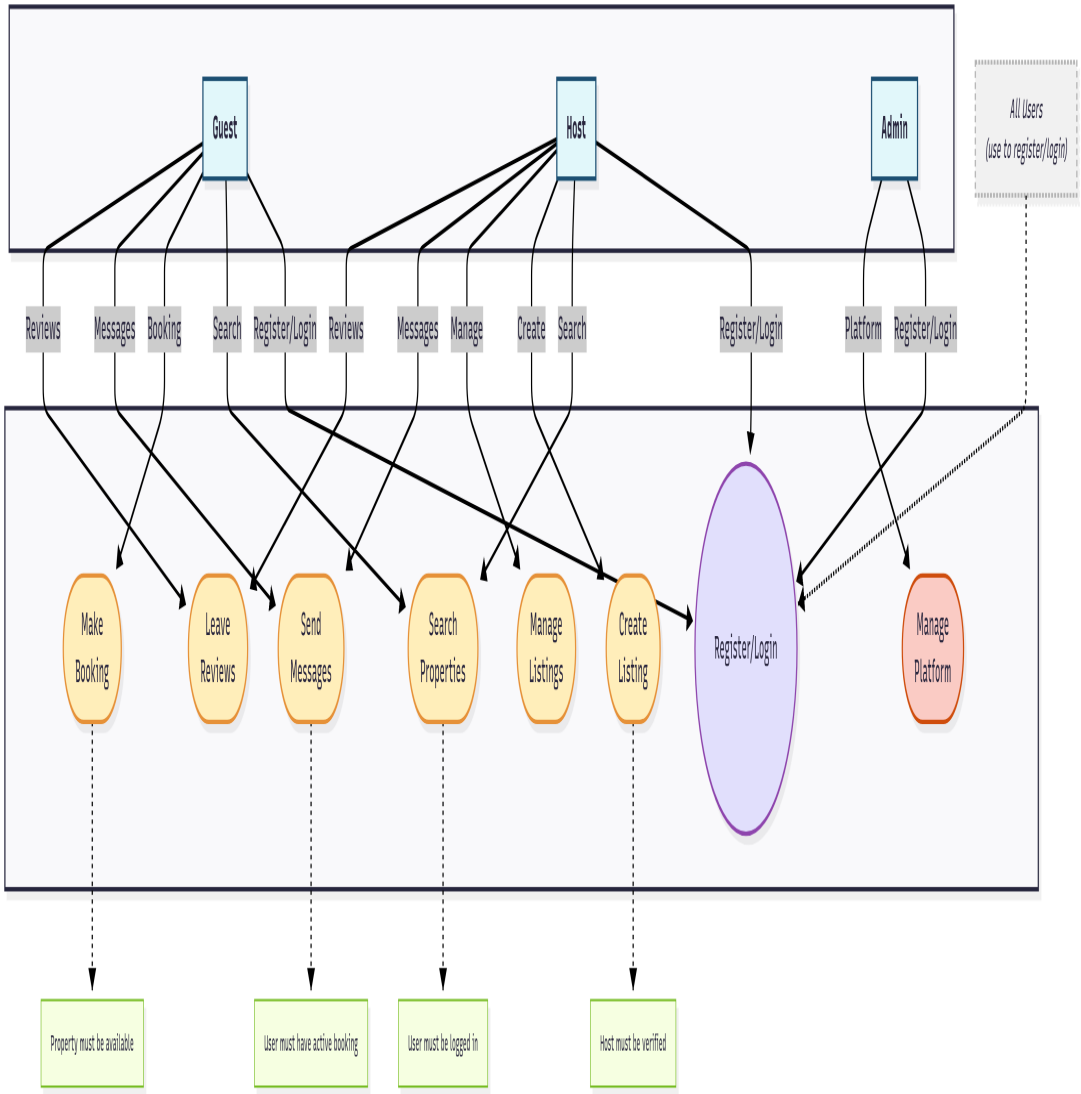

Figure 1: Use Case Diagram showing comprehensive interactions between Guest, Host, Admin and Otithi Platform

5.2 Use Case Narrative Table

Use Case	Actor	Type	Description	Precondition
Register/Login	All Users	Primary	User account creation and authentication with security measures	None
Search Properties	Guest	Primary	Find available accommodations with advanced filtering	User logged in
Create Listing	Host	Primary	Add property to platform with verification workflow	Host account verified
Make Booking	Guest	Primary	Reserve property for specific dates with payment	Property available
Manage Listings	Host	Secondary	Edit property details and manage availability	Host owns listing
Send Messages	Host/Guest	Secondary	Communicate with other users securely	Active booking
Leave Reviews	Guest	Secondary	Rate and review property after stay	Completed stay
Manage Platform	Admin	Primary	Oversee platform operations and users	Admin privileges
Process Payments	System	Primary	Handle financial transactions securely	Valid booking
Verify Users	Admin	Secondary	Review and approve user accounts	Admin access

Table 10: Comprehensive Use Case Narrative Table

5.3 Data Flow Diagram

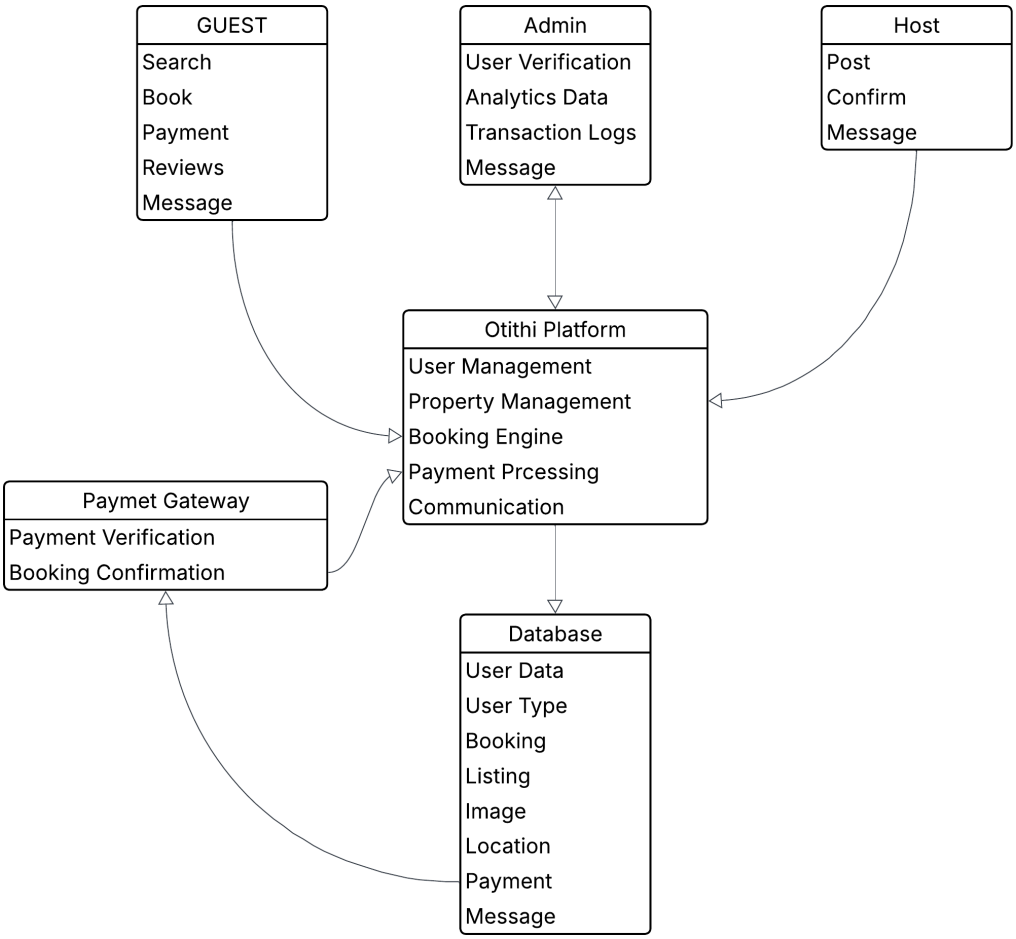

Figure 2: Data Flow Diagram (Level 0) showing system context and external entities

5.4 Data Flow Diagram - System Decomposition

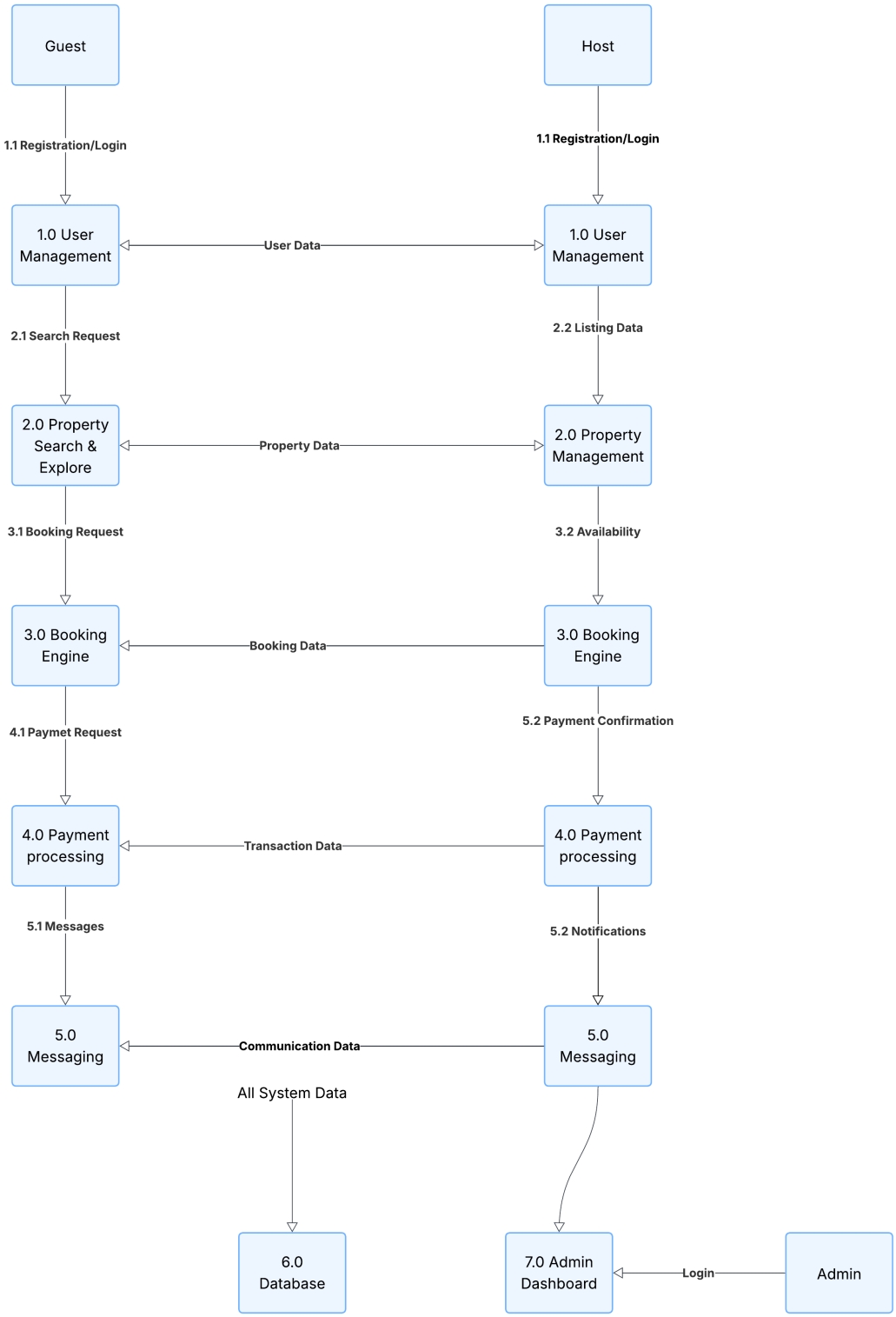

Figure 3: Data Flow Diagram (Level 1) showing System Decomposition and internal processes

5.5 Class Diagram

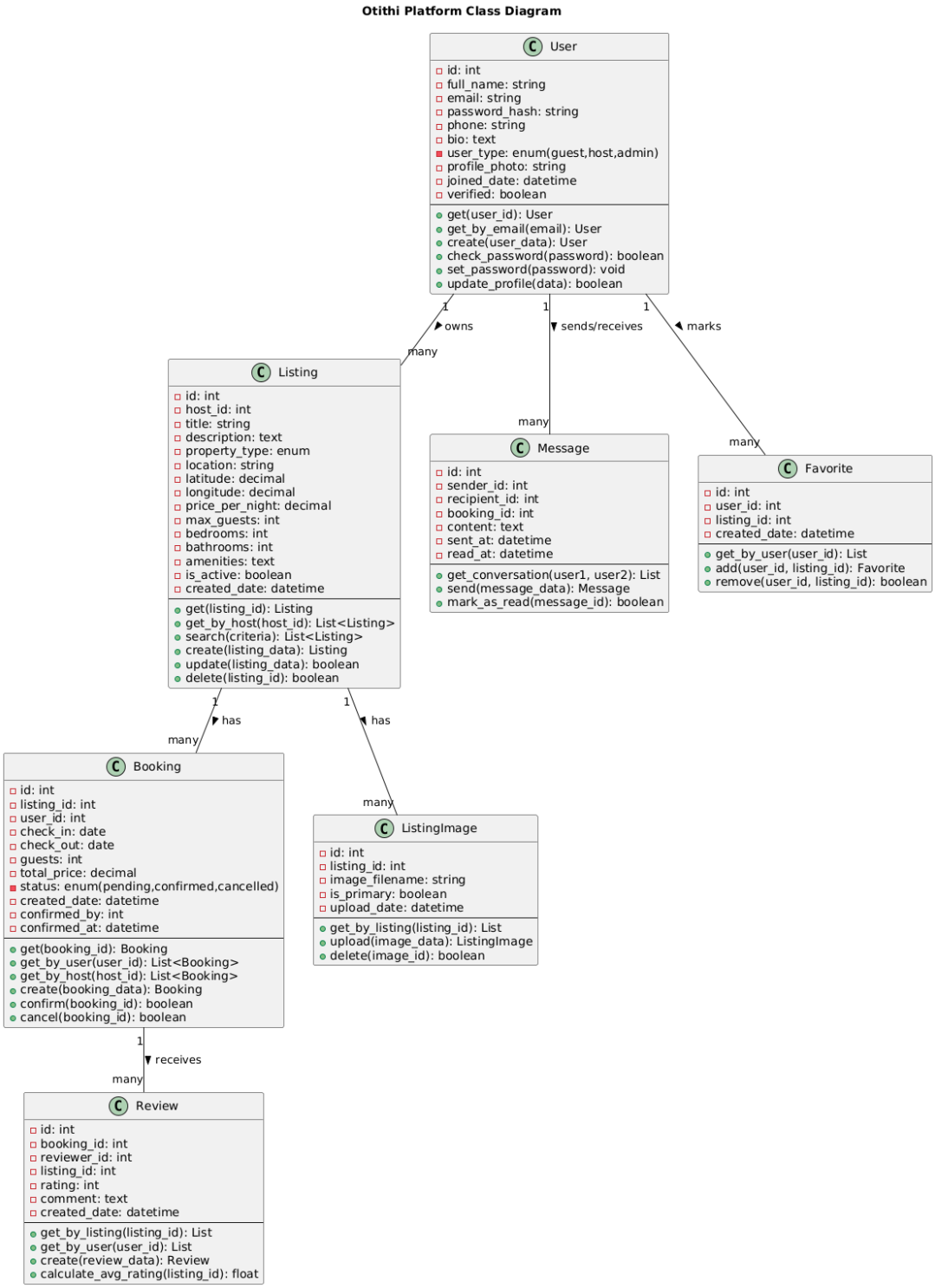

Figure 4: Class Diagram showing main system entities, relationships, and data structures

5.6 Activity Diagram

Activity ID	Activity	Description	Dependencies
1	Start	User accesses platform	None
2	User Authentication	Account creation or authentication	None
2.1	Register	Create new account	None
2.2	Login	Access existing account	None
3	Property Discovery	Find available accommodations	2
3.1	Browse Properties	View available listings	2
3.2	Search & Filter	Apply search criteria	3.1
3.3	View Property Details	Examine specific listing	3.2
4	Property Management	Manage property listings	2
4.1	Create Listing	Add new property	2
4.2	Edit Listing	Modify existing properties	4.1
4.3	Manage Availability	Control booking dates	4.1
5	Booking Process	Reserve accommodation	3.3
5.1	Check Availability	Verify booking dates	5
5.2	Make Reservation	Confirm booking	5.1
5.3	Process Payment	Complete financial transaction	5.2
6	Communication	Interact with other party	5.2
6.1	Send Messages	Communicate securely	6
6.2	Receive Notifications	Get platform updates	6
7	Post-Stay	Complete stay experience	5.3
7.1	Leave Review	Rate and review experience	7
7.2	Host Response	Reply to guest feedback	7.1
8	Admin Management	Oversee platform operations	2
8.1	User Verification	Approve user accounts	8
8.2	Content Moderation	Monitor platform content	8
8.3	Analytics Review	Analyze platform performance	8
9	End	Complete user session	All activities

Table 11: Activity Flow Description Table with Hierarchical Dependencies

6 Non-Functional Requirements

The Otithi platform must meet comprehensive non-functional requirements across multiple dimensions to ensure user satisfaction, system reliability, and business success. These requirements are categorized into eight key areas:

Performance Requirements:

1. **NFR-01:** Page load time must not exceed 3 seconds under normal load conditions, ensuring responsive user experience across all devices and network conditions.
2. **NFR-02:** System must support 1,000 concurrent users without performance degradation, maintaining consistent response times and user experience quality.
3. **NFR-03:** Database queries must complete within 1 second for 95% of requests, ensuring efficient data retrieval and system responsiveness.
4. **NFR-04:** Image upload and processing must complete within 10 seconds for files up to 5MB, supporting efficient content management for hosts.

Security Requirements:

1. **NFR-05:** Strong password policy enforcement requiring minimum 8 characters with mixed case, numbers, and special characters, ensuring account security.
2. **NFR-06:** All passwords must be hashed using industry-standard algorithms (bcrypt) with appropriate salt rounds, protecting user credentials even in case of data breach.
3. **NFR-07:** Session timeout after 60 minutes of inactivity, automatically logging out users to prevent unauthorized access to accounts.
4. **NFR-08:** CSRF protection on all state-changing operations, preventing cross-site request forgery attacks and ensuring data integrity.
5. **NFR-09:** HTTPS/TLS encryption for all data transmission, protecting sensitive information including payment details and personal data.

Usability Requirements:

1. **NFR-10:** Mobile-responsive design supporting devices with screen widths from 320px and above, ensuring optimal experience across all devices.
2. **NFR-11:** Intuitive navigation with maximum 3 clicks to access any feature, providing efficient user workflows and reducing user frustration.
3. **NFR-12:** Bengali language support for key interface elements, accommodating local language preferences and improving accessibility.
4. **NFR-13:** Accessibility compliance with WCAG 2.1 AA standards, ensuring the platform is usable by people with disabilities.

Reliability Requirements:

1. **NFR-14:** System uptime of 99.5% (maximum 36 hours downtime per year), ensuring consistent availability for users and maintaining business continuity.
2. **NFR-15:** Automated daily backups with 30-day retention, protecting against data loss and enabling disaster recovery scenarios.
3. **NFR-16:** Graceful error handling with user-friendly error messages, preventing system crashes and providing helpful guidance to users.

Scalability Requirements:

1. **NFR-17:** Horizontal scaling capability to support 10x user growth without architectural changes, ensuring long-term platform sustainability.
2. **NFR-18:** Database optimization for handling 100,000+ property listings, maintaining performance as the platform grows.

Compatibility Requirements:

1. **NFR-19:** Cross-browser compatibility with Chrome, Firefox, Safari, and Edge (latest versions), ensuring consistent experience across different browsers.
2. **NFR-20:** Mobile device compatibility with iOS 12+ and Android 8+, covering the majority of mobile users in Bangladesh.

Maintainability Requirements:

1. **NFR-21:** Modular code architecture enabling easy feature additions and modifications, supporting rapid development and maintenance.
2. **NFR-22:** Comprehensive logging and monitoring systems for proactive issue identification and resolution.

Compliance Requirements:

1. **NFR-23:** Compliance with Bangladesh data protection regulations, ensuring legal compliance and user privacy protection.
2. **NFR-24:** Payment card industry (PCI) compliance for secure payment processing, protecting financial information and maintaining trust.

7 Risk Analysis

7.1 SWOT Analysis

STRENGTHS
• Local market focus specifically designed for Bangladesh's unique needs and cultural context • Cultural alignment with Bengali language support and traditional hospitality values • Modern, robust technology stack (Flask, MySQL, Bootstrap) ensuring reliability and scalability • Comprehensive feature set with BDT integration addressing local market requirements • User-centric design philosophy with mobile optimization for Bangladesh's mobile-first market • Strong understanding of local regulatory environment and business practices

Table 12: SWOT Analysis - Strengths

WEAKNESSES
• New brand without established market presence or recognition in the accommodation sector • Limited financial and human resources compared to established international competitors • Dependency on local payment solutions and banking infrastructure for financial operations • Initial challenge in generating sufficient content and listings to attract users • Need to establish comprehensive customer support infrastructure and processes • Limited experience in the accommodation sharing market compared to established players

Table 13: SWOT Analysis - Weaknesses

OPPORTUNITIES
• Rapidly growing Bangladesh tourism industry with 15-20% annual growth rates • Increasing digital adoption and smartphone usage creating tech-savvy user base • Government initiatives promoting tourism development and digital transformation • Partnership opportunities with hotels, travel agencies, and tourism boards • Mobile-first market with high growth potential and limited competition • Growing middle class with increasing disposable income for travel and accommodation

Table 14: SWOT Analysis - Opportunities

THREATS
• Entry of established international competitors with significant resources and brand recognition • Potential regulatory changes affecting short-term rentals and accommodation sharing • Economic uncertainty and currency fluctuations impacting travel demand and spending • Cybersecurity threats and potential data breaches affecting user trust and platform security • Initial user trust and safety concerns requiring significant effort to overcome • Competition from traditional accommodation providers adapting to digital platforms

Table 15: SWOT Analysis - Threats

7.2 Risk Mitigation Strategies

The platform implements comprehensive risk mitigation strategies across multiple dimensions:

Technical Risks: Database performance issues are mitigated through strategic indexing, query optimization, and caching mechanisms. Security vulnerabilities are addressed through regular security audits, penetration testing, and continuous monitoring systems. Payment gateway failures are mitigated through multiple provider integration and fallback systems.

Operational Risks: Server downtime risks are minimized through load balancing, comprehensive monitoring, and automated failover systems. Mobile compatibility issues are addressed through comprehensive cross-device testing and responsive design principles.

Business Risks: Market competition risks are mitigated through strong local market focus and cultural alignment. Regulatory risks are addressed through proactive engagement with relevant authorities and flexible system architecture.

Financial Risks: Currency fluctuation risks are managed through local currency focus and strategic pricing strategies. Economic downturn risks are mitigated through diversified user base and flexible business models.

8 Conclusion

The Otithi platform represents a comprehensive, well-architected solution for the accommodation rental market in Bangladesh. By combining modern web technologies with deep understanding of local market needs, cultural context, and user preferences, the platform addresses significant gaps in the current accommodation booking ecosystem while positioning itself for sustainable growth and market leadership.

The name "Otithi" (*otithi*) embodies the traditional Bangladeshi spirit of hospitality and the cultural value of treating guests with respect and warmth. This cultural alignment makes the platform not only functionally superior but also emotionally resonant with local users, while remaining accessible and appealing to international visitors seeking authentic experiences.

Expected Benefits and Impact:

The platform delivers comprehensive benefits across multiple stakeholder groups:

For Guests: Access to diverse, authentic accommodation options with secure booking processes in local currency (BDT), enhanced travel experiences through local host connections, and cost-effective alternatives to traditional accommodation options.

For Hosts: New revenue streams through property monetization, opportunities to showcase local hospitality and culture, and participation in the growing digital economy with minimal technical barriers.

For Bangladesh: Tourism growth and digital advancement, economic impact through revenue generation and employment opportunities, and enhanced international reputation as a welcoming destination.

Key Success Factors:

The platform's success depends on several critical factors:

User Experience: Intuitive, responsive design that encourages adoption and continued use across all devices and user types.

Security and Trust: Robust verification systems, comprehensive security measures, and transparent review mechanisms that build and maintain user trust.

Local Market Alignment: Features and functionality specifically tailored to Bangladesh market needs, including BDT integration, Bengali language support, and cultural context.

Technical Excellence: Scalable, maintainable architecture that supports growth and enables rapid feature development and deployment.

Community Building: Fostering positive host-guest relationships, promoting cultural exchange, and building a trusted community of users.

The comprehensive requirements specification, technical architecture, risk mitigation strategies, and cultural alignment provide a solid foundation for successful implementation of a platform that meets user needs, business objectives, and contributes to Bangladesh's digital transformation and tourism development goals.

9 Abbreviations

Abbreviation	Full Form
API	Application Programming Interface
UI	User Interface
UX	User Experience
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
HTTP	HyperText Transfer Protocol
HTTPS	HyperText Transfer Protocol Secure
CSRF	Cross-Site Request Forgery
SSL	Secure Sockets Layer
WBS	Work Breakdown Structure
WCAG	Web Content Accessibility Guidelines
PCI	Payment Card Industry
OWASP	Open Web Application Security Project
ICT	Information and Communication Technology
AI	Artificial Intelligence
REST	Representational State Transfer
CRUD	Create, Read, Update, Delete
DBMS	Database Management System
SQL	Structured Query Language
JSON	JavaScript Object Notation
XML	eXtensible Markup Language
URL	Uniform Resource Locator

Table 16: List of Abbreviations Used in the Document

10 References

10.1 Technical References

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6. Python Documentation: https://docs.python.org/ - Programming language reference and development guidelines
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8. CSS3 Documentation: https://www.w3.org/Style/CSS/ - Styling language specifications and best practices

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10.3 Academic and Research References

1. Software Engineering Institute: Requirements engineering best practices and software specification standards
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